

Aruvi — the external-provider seam

Every vendor-shaped dependency sits behind a named interface. This sheet shows each slot, what fills it today, and what a production partner is being asked to supply.

STATUS

● Working plug ● Stub ● Seam bypassed ● Reserved

ARUVI'S OWN CODE — RETAINED, NOT REBUILT

Answers Section 3 of the Technology Partnership & Production Roadmap: what can be retained, and what must be replaced.

web/ <i>Next.js</i> The teacher-facing app. Mobile-first and responsive. Speaks HTTP to the API and to nothing else.	api/ <i>FastAPI service</i> The service layer. Every provider in the product is chosen in ONE block: main.py, lines 79-214.	aruvi_core/ <i>the engine</i> Lesson generation, period allocation, canonical view model, five subject plugins. Pure Python, no UI.	genon/ <i>authoring pipeline</i> Runs offline to build the lesson library. Not part of the live service, and not part of the handover.
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No vendor name appears anywhere above this line.

aruvi_core/ports.py — 22 named interfaces

The engine depends on these and never on a provider. Swapping a vendor is one new adapter class plus one line in api/main.py.

T H E S E A M

EXTERNAL PROVIDERS — THE SLOTS A PRODUCTION PARTNER FILLS

Three have working plugs, two are stubs, two are reserved for later — and one seam is not yet honoured.

WORKING PLUG · FILES Database / teacher state 12 × <code>..Repository</code> TODAY Twelve file adapters under data/cloud/state/. Every method already takes <code>tenant_id</code> and <code>user_id</code> (teaching state adds <code>year_id</code>) and paths are keyed to match — so tenant isolation is already in the shape of the data, not a later retrofit. PARTNER SUPPLIES Managed Postgres (e.g. Supabase) behind the same twelve ports, plus row-level security.	STUB · THE SECURITY GAP Authentication <code>AuthProvider</code> TODAY <code>HeaderAuthProvider</code>: the user ID arrives in an X-Aruvi-User header and there is no password. Development only. Exactly one caller, <code>_current_identity()</code>. PARTNER SUPPLIES A real identity provider with token verification. Replaces one class and one line.	STUB · MANUAL TODAY Payments <code>BillingProvider</code> TODAY <code>ManualBillingProvider</code> — the founder grants entitlement by CLI. Everything around it is built: entitlement scopes, invoice series, consent record, priced cart. PARTNER SUPPLIES An Indian gateway plus a webhook route, and advice on Play / App Store fees across web and mobile.	WORKING PLUG · REAL SMTP Transactional email <code>Notifier</code> TODAY <code>SmtplibNotifier</code> sends genuinely once credentials are set in the environment; a file outbox stands in otherwise. The choice is made once, at the seam, on startup. PARTNER SUPPLIES Optional — a deliverability provider if volume warrants it. Nothing here is blocking.
SEAM BYPASSED · REFACTOR Content / object store <code>Storage</code> (declared, bypassed) TODAY The one seam not yet honoured. The lesson library is read straight off the filesystem — roughly fourteen <code>open()</code> / <code>json.load()</code> calls sit in <code>api/data.py</code> rather than behind the port. PARTNER SUPPLIES Route those reads through the <code>Storage</code> port, then S3 / GCS / Supabase Storage behind it.	RESERVED · NONE AT RUNTIME AI / LLM <code>LLMClient</code>, <code>OutputCache</code> TODAY Not wired, deliberately. The live service calls no AI at all — it serves pre-authored plans by deterministic selection. There is no AI key in production to leak. PARTNER SUPPLIES Nothing now. An adapter only if live generation is later wanted.	RESERVED · UNUSED Background jobs <code>JobQueue</code> TODAY Declared and unused. Nothing in the product currently needs asynchronous work; exports render inline. PARTNER SUPPLIES Nothing now. Relevant only if batch work later moves server-side.	WORKING PLUG · BUILT Data rights (DPDP · Apple 5.1.1(v)) <code>DataRightsService</code> TODAY Export walks every state store into a single Word file; erase returns a signed receipt. Neither is ever gated on subscription state. PARTNER SUPPLIES Nothing of its own — it travels with the database adapter.

Config, not code

Every provider setting is an environment variable in `api/config.py`. No secret is committed to the repository.

The contracts are tested

42 test files exercise the ports, so a swapped adapter is verifiable rather than hopeful.

Ownership

Cloud, database, domain, store and gateway accounts to be created in Aruvi's own name. The partner receives access, not ownership.